Varshini Bhavanam

Software Engineer

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Summary

Software engineer with production full-stack experience across Python, TypeScript/React, and Ruby — built and shipped analytics pipelines, automation systems, and customer-facing web platforms handling tens of thousands of records in a live 7-state operation. Comfortable across the stack: backend APIs, database design, background job processing, frontend dashboards, and self-hosted Linux infrastructure.

Education

University of Minnesota Duluth — Duluth, MN
Bachelor of Science in Computer Science — *December 2025*

Technical Skills

- **Languages:** Python, TypeScript/JavaScript, Ruby, SQL
- **Frontend:** React 19, Next.js, Angular, Vite, Tailwind CSS, Supabase (Realtime, Auth, Edge Functions)
- **Backend:** Ruby on Rails, Flask, ASP.NET Core, REST APIs, PostgreSQL, MongoDB, Airtable, Neon
- **Testing:** RSpec, factory_bot, WebMock/VCR, Jest, pytest
- **DevOps & Infra:** Docker, GitHub Actions (CI/CD), Terraform, AWS (VPC, ALB, ASG, RDS, ECR, CloudWatch), Kubernetes, Ansible, Cloudflare Pages/Workers/Tunnel, Proxmox, Caddy, systemd, Kamal, SolidQueue, Sidekiq
- **Integrations:** Stripe, OptimoRoute, DocuSign, RingCentral, Odoo, Microsoft Graph API
- **Collaboration:** Jira, Agile/Scrum, Power BI, Tableau, Streamlit
- **Business Systems:** Vonage, RingCentral, Mailchimp, Meta Pixel, Google Analytics 4, SEMrush, Django

Experience

Xecutive Development LLC — Feb 2026 – Present
Software Developer — *Minneapolis, MN*

- Built a Rails 8 + PostgreSQL (Neon) production service with SolidQueue background jobs that syncs Airtable customer data nightly across 7 OptimoRoute state accounts, exposing a bearer-token-authenticated internal API that other apps in the suite query for live customer lookups — eliminating hours of manual data entry and serving as the company's single source of truth.
- Engineered AirBridge, a Python payment-reconciliation pipeline: ingests CSV payment exports, joins Stripe customer and payment-method IDs via email key, segments customers for renewal targeting, and writes results back to the Airtable CRM with append-only history and dry-run/preview safety gates so a production sync never silently overwrites multi-year enrollment data.
- Provisioned and deployed the production service on a self-managed Proxmox VM, configuring Caddy as an automatic-HTTPS reverse proxy and Cloudflare Tunnel for zero-open-port ingress, with systemd-managed deploys, a self-hosted GitHub Actions runner for CI/CD, and Brakeman/bundler-audit security scans gating every merge.
- Built a React 19 + TypeScript + Supabase real-time inventory tracker deployed on Cloudflare Pages — role-based access control and QR-code equipment scanning across multiple warehouse locations, with Supabase Realtime pushing check-out/check-in updates instantly to every connected dashboard.
- Authored Jupyter-based ETL notebooks to migrate 6,258 contacts, 3,448 sales orders, and 3,299 payments from Airtable into a new Odoo ERP, including data-cleaning, deduplication, and external-ID mapping via Odoo's XML-RPC API.
- Pioneered an AI-powered SMS hiring assistant (Claude + RingCentral) that qualifies and onboards field operators end-to-end, auto-triggering DocuSign agreements and CRM status updates with zero human touch.

- Redesigned the internal operator-management dashboard (React + TypeScript) with a token-based light/dark theming system and a KPI/retainer-chart layout, and fixed a stale-closure race condition in the inventory tracker that was causing duplicate equipment rentals.
- Built a **multi-state sales-tax filing app** (Django + PostgreSQL, Docker) for MN, CO, and NY — state tax APIs, ZIP+4 geocoding, and a 130-column Avalara-format export; a 16-worker pipeline cut a 1,021-row batch from ∼340s to **11.9s**.
- Took the **Stripe renewal engine** to production — dry-run mode with zero Stripe calls, Lawn/Snow season split, a database-driven email-template editor with live preview, and tests grown from 110 to 129 examples.
- Hardened authentication on the internal app launcher (Next.js + Auth.js) — cleared **3 critical and 3 high** security advisories, fixed a fail-open auth check, and added login rate limiting and timing-safe credential checks.

Code Ninjas Jul 2025 – Sep 2025
Software Development Intern *Atlanta, GA*

- Designed and delivered a full-stack AI quiz-generation platform using Angular and ASP.NET Core; integrated NLP APIs for automated quiz creation from natural-language input — reduced manual content-creation time by **80%**.
- Built a rule-based Python chatbot walking students through curriculum problems to earn skill badges; containerized with Docker and deployed via GitHub Actions CI/CD.

University of Minnesota Sep 2024 – May 2025
Undergraduate Research Assistant *Duluth, MN*

- Compiled IRB protocol and consent documentation for a mobile wellness study; analyzed user-behavior data and co-authored two peer-reviewed publications.

St. Luke's Aspirus Administration Feb 2025 – Apr 2025
EHR Data Migration Intern *Duluth, MN*

- Transferred and validated 10,000+ patient records from Meditech to Epic during an enterprise EHR cutover; identified and documented data discrepancies under HIPAA-compliant workflows.

Selected Projects

AI-Model-Stock-Analysis — *Python, FastAPI, Next.js, XGBoost, PyTorch (LSTM), Docker, Fly.io*

- Built a FastAPI backend serving both a Streamlit and a Next.js/TypeScript frontend from one shared codebase — no duplicated business logic across UIs.
- Trained an XGBoost/LSTM ensemble on price, sentiment (FinBERT/VADER), and macro features with no-look-ahead discipline; validated with walk-forward backtesting including realistic transaction costs and slippage.
- Shipped portfolio-risk tooling (correlation-regime detection, drawdown tracking, tax-loss-harvesting scanner) and deployed via Docker to Fly.io with GitHub Actions CI across three Python versions.

Production AWS Deployment Pipeline — *Terraform, AWS, Docker, Kubernetes, Ansible*

- Wrote Terraform provisioning a full AWS network + compute + data tier (VPC across 2 AZs, ALB, Auto Scaling Group, RDS PostgreSQL, ECR) with least-privilege security-group chaining and CloudWatch alarm-driven autoscaling.
- Built a 4-stage GitHub Actions pipeline (lint → test → build/push to ECR → Ansible-driven EC2 deploy) with an alternate Kubernetes manifest path and Slack-alerting CloudWatch dashboard.

Publications

- Noshin, A., Bhavanam, V., et al. "Humanoid and Socially Assistive Robots in Mental Health Care." *ICICT 2025.*
- Ferdousee, Z., Bhavanam, V., et al. "Digital Interventions for Depression and Anxiety." *HEALTHINF 2026.*

Certifications

- Microsoft Business Analyst Professional Certificate — Microsoft via Coursera, October 2025
- Google Cybersecurity Professional Certificate — Google via Coursera, May 2025